

Onyero Walter Ofuzim

Calgary, AB, Canada

✉ Email

🐙 GitHub

🌐 Portfolio

🌐 LinkedIn

🔍 ResearchGate

+1 403 605 0955

Research Interests

Embedded and IoT security, systems security, firmware analysis, software testing and fuzzing, access control for connected systems, secure embedded software, program analysis, and dependable cyber-physical systems.

Education

University of Calgary

Calgary, AB, Canada

M.Sc., Electrical & Software Engineering

Jan 2024 – May 2026 (expected)

GPA: 3.85

Focus: control systems, system identification, optimization, battery modelling, machine learning for dynamical systems

- Department of Electrical & Software Engineering Funding 2024
- International Graduate Tuition Award 2024
- Dept./Faculty Scholarship Award (Exceptional Student) 2024

University of Benin

Benin City, Nigeria

B.Eng., Electrical / Electronic Engineering

2021

Focus: embedded systems, instrumentation, control systems, electronics

Academic Research Experience

Graduate Research Assistant, University of Calgary

Jan 2024 – Present

Battery Modeling and System Identification Research

- Developed a modular Python research codebase for structured modelling, parameter estimation, automated experiment workflows, and reproducible evaluation.
- Built multi-stage estimation and validation pipelines for nonlinear dynamical systems using Python, JAX, SciPy, and test-driven scientific workflows.
- Designed automated preprocessing, cycle-detection, and analysis tooling for large experimental datasets with emphasis on maintainability, traceability, and reproducibility.
- Produced technical documentation, experiment logs, and structured reports for supervisor and project stakeholders.

Graduate Teaching Assistant, University of Calgary

Jan 2024 – Present

- Supported courses in software engineering practices, computer organization, controls, optimization, analog electronics, and electric machines/drives.
- Mentored students on Git, testing, debugging, structured programming, and reproducible technical workflows.

Graduate Research Assistant, University of Benin

Jan 2020 – Aug 2021

Embedded Systems and IoT for Energy Applications

- Prototyped a GSM-enabled battery monitoring and control device for remote supervision of distributed energy assets.
- Worked with embedded sensing, control logic, and communication interfaces for IoT-enabled monitoring systems.
- Co-authored a publication on remote battery monitoring and control using IoT-enabled embedded hardware.

Engineering and Systems Experience

MTN Nigeria, Network Performance / Support Roles

Nov 2019 – Nov 2023

- Built Python-based automations and KPI analysis workflows for operational network monitoring and troubleshooting.
- Worked with Linux terminals, SNMP-based monitoring environments, incident reporting, and service-performance diagnostics.
- Supported troubleshooting across IP/MPLS, data, internet, and operational support environments, developing practical systems and networking intuition relevant to connected-device ecosystems.

Automation / Electrical Engineer Intern, Nigerian Bottling Company

Aug 2019 – Nov 2019

- Supported troubleshooting, inspection, and maintenance documentation for industrial electrical and automation systems.

Selected Projects Relevant to Embedded / Systems / Security Transition

- **Cisco-Style Multi-Client TCP Echo Server** — Implemented a C / BSD sockets networking project with `poll()`, partial read/write handling, line-based protocol logic, and multi-client concurrency. [GitHub]
- **API Smoke Tests — reqres.in** — Built a Python-based API testing suite with retries, negative cases, reporting, and CI-friendly execution. [GitHub]
- **UI E2E — SauceDemo** — Developed end-to-end automated test flows using Selenium, Behave, Page Object Model design, and GitHub Actions. [GitHub]
- **High-Performance Systems & Algorithms Bench** — Built a benchmarking framework combining modern C++ kernels and Python experiment orchestration for repeatable systems testing and performance analysis. [GitHub]
- **Battery Monitoring & Control Device using IoT** — Designed and implemented an embedded remote battery monitoring and control prototype using PIC16F887 and GSM/GPRS communication. [ResearchGate]
- **Physics-Informed Battery Degradation Monitoring using SPMe System Identification** — Developed a research software package with modular estimation, validation, and testing workflows for nonlinear dynamical systems. [GitHub]

Publications and Research Outputs

- **Ofuzim, O. W.**, Leal, D. *Proton Exchange Membrane Electrolyzer Battery Data Prediction Using NARX (Nonlinear AutoRegressive with Exogenous Inputs) Neural Network Model*. Available via ResearchGate.
- **Ofuzim, O. W.** (Co-author). *Kinematic Analysis of Omnidirectional Mecanum Wheeled Robot*. [ResearchGate]
- **Ofuzim, O. W.** (Co-author). *Design and Construction of a Remote Battery Monitoring and Control Device Using the Internet of Things*. [ResearchGate]

Technical Skills

Programming: Python, C/C++, MATLAB, Bash, SQL, JavaScript/TypeScript

Systems / Software: Linux, Git/GitHub, pytest, unittest, Selenium, CI workflows, debugging, technical documentation

Networking / Embedded: TCP/IP, BSD sockets, SNMP monitoring, UART/I²C/CAN concepts, GSM-enabled embedded IoT prototyping

Security-Relevant Foundations: software testing, automation, reproducible experimentation, systems programming, protocol handling, secure-systems interest

Control / Modeling: state-space methods, observers, Kalman filtering, system identification, optimization, nonlinear modelling

Certifications (Selected)

- Open Source Software Development, Linux and Git Specialization — Linux Foundation
- Programming in Python — Meta
- Programming with JavaScript — Meta
- Data Engineer in Python — DataCamp
- Machine Learning — Stanford Online

Professional Notes

Available to start PhD studies after completion of current M.Sc. degree requirements. Motivated to transition into embedded and IoT security through research in firmware analysis, fuzzing, and secure access control systems.